

AP Chem summer homework

2025-07-31

2 Math problems

- a. 1.81×10^6
- b. 1.43×10^6
- c. 2.482×10^{-7}
- d. -1.738×10^{-7}
- e. 6.2×10^8
- f. 2×10^{-3}
- g. 5
- h. 0.3
- i. 2×10^{-5}
- j. $\pm 16/15$
- k. 4×10^{-5}
- l. 4
- m. -10.68
- n.

$$x^2 + 0.1x = 2 \times 10^{-8}$$
$$x^2 + 0.1x - 2 \times 10^{-8} = 0$$

$$x = \boxed{2 \times 10^{-7}}$$

o.

$$x + y = 3$$

$$x - y = 9$$

$$2x = 12$$

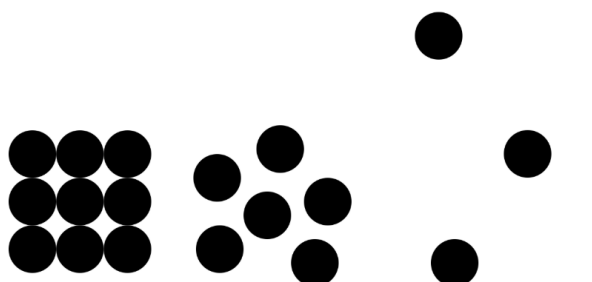
$$\boxed{x = 6}$$

3 Significant figures

- a. 5
- b. 3
- c. 4
- d. 3

- e. 5
- f. 1
- g. 1
- h. 6
- i. 5
- j. 6

4 Chapter 1



solid	liquid	gas
• not compressible • fixed shape • fixed volume	• slightly compressible • no fixed shape • fixed volume	• compressible • no fixed shape • no fixed volume

- a.
- b. iii
- c. 19%
- d. Dimensional analysis-
 - i.

$$0.069 \text{ nm} = 6.9 \times 10^{-9} \text{ cm}$$

$$V = \frac{4}{3}\pi(6.9 \times 10^{-9})^3 = 1.38 \times 10^{-24} \text{ cm}^3$$

$$\frac{3.35 \times 10^{-23} \text{ g}}{1.38 \times 10^{-24} \text{ cm}^3} = 24.3 \text{ g/cm}^3$$

- ii. 10^3 ; 10^2 ; 10^{-3} ; 10^{-6} ; 10^{-9}

- iii.

$$\frac{330 \text{ minutes}}{1 \text{ day}} \cdot \frac{1 \text{ hour}}{60 \text{ minutes}} = 5.5 \text{ hours/day}$$

iv.

$$1 \text{ year} \times \frac{365 \text{ days}}{1 \text{ year}} \times \frac{24 \text{ hours}}{1 \text{ day}} \times \frac{60 \text{ minutes}}{1 \text{ hour}} \times \frac{60 \text{ seconds}}{1 \text{ minute}} = 3.1536 \times 10^7 \text{ seconds}$$

v.

$$10,000 \text{ L} \times \frac{1000 \text{ ml}}{1 \text{ L}} \times \frac{1 \text{ cm}^3}{1 \text{ ml}} = 10^7 \text{ cm}^3$$

vi.

$$\frac{1.2 \times 10^9 \text{ gal}}{1 \text{ day}} \times \frac{1 \text{ day}}{24 \text{ hours}} \times \frac{1 \text{ hour}}{60 \text{ minutes}} \times \frac{1 \text{ minute}}{60 \text{ seconds}} = 13888.88 \text{ gallons/second}$$

$$\frac{1 \text{ day}}{1.2 \times 10^9 \text{ gal}} \times \frac{1.3 \times 10^{15} \text{ gal}}{=} 1.08 \times 10^6 \text{ days}$$

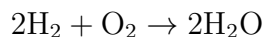
vii.

$$2.5 \text{ pounds} \times \frac{453,000 \text{ scales}}{1 \text{ pound}} \times \frac{12 \text{ hours}}{8 \text{ scales}} \times \frac{1 \text{ bee}}{12 \text{ hours}} \approx 10,108 \text{ bees}$$

- e. Element: oxygen; compound: water; homogeneous mixture: dish soap; heterogeneous mixture: oil in water; solution: saltwater

5 Chapter 2

- a. The mass number is simply the sum of protons plus neutrons in an atom, while average atomic mass is the average of all the masses of all isotopes of an element.
- b. Mass can neither be created nor destroyed
- c.



Available: 4 moles H_2 (2 g/mol) and 1 mole of O_2 (32 g/mol); O_2 is the limiting reactant

Molar mass of H_2O = 18 g/mol

1 : 2 ratio between O_2 and H_2O

$$2 \times 18 = 36 \text{ g } \text{H}_2\text{O}$$

- d. Molar mass of K_2SO_4 : 174.26 g/mol

$$\text{K}_2 - 2 \cdot 39.1 = 78.2 \text{ g/mol}$$

$$\text{S} - 32.06 \text{ g/mol}$$

$$\text{O}_4 - 4 \cdot 16 = 64 \text{ g/mol}$$

Assuming 1 mol, K makes up 78.2 of 174.26 grams

$$\frac{78.2}{174.26} \times 100\% = \boxed{44.88\%}$$

- e. Nucleus: center of the atom consisting of protons and neutrons; orbitals: areas where electrons are most likely to be found, organised by energy level; electrons: particles with negligible mass that are found in the orbitals surrounding the nucleus, adding negative charge; neutrons: a particle found in the nucleus with neutral charge holding protons together
- f. Atoms with the same number of protons but different numbers of neutrons (same charge, different mass)
- g. C-14
- h. Ionic bonding is the bonding of atoms through electrostatic attraction, a result of one atom taking electrons from another to fill both their valence shells; covalent bonding is the bonding of atoms through the sharing of electrons.
- i. A chemical formula shows the number of each type of atom in a compound, and a structural formula creates a visual to show where the atoms are connected to each other in space.
- j. ion: an atom with a positive or negative net charge due to an unequal number of protons and electrons; cation: an ion with positive charge; anion: an ion with negative charge.
- k. Metals are characterised by high conductivity, lustre, and malleability. They lose electrons to nonmetals in ionic bonding. Nonmetals are found on the upper right side of the periodic table, and they take electrons during ionic bonding. They are dull, brittle, and poor conductors of heat and electricity. They also form covalent bonds with each other. Metalloids are elements that fall in between these two sets of characteristics. They are semiconductors and can form both ionic and covalent bonds.
- l. Periods are the rows, and groups are the columns on the periodic table.
- m. The s-block, p-block, and d-block are the groups that the periodic table is divided into based on the highest energy electron orbital.
- n. Group 1, Group 2, Group 7, Group 8
- o.
 - i. Sodium chloride, calcium fluoride, lithium oxide, iron (III) oxide, silver (I) fluoride
 - ii. MgSe, RbI, Na₃P, Cu₂S, HgO
 - iii. Ammonium chloride, sodium hydroxide, silver nitrate, manganese (II) carbonate
 - iv. KBrO₃, Fe(NO₃)₃
 - v. Phosphorus trichloride, phosphorus pentachloride, sulfur dioxide, carbon dioxide
 - vi. SO₃, SF₆, O₂F₂
 - vii. Hydrochloric acid, acetic acid, sulfuric acid
 - viii. H₃PO₄, HNO₃, HNO₂

6 Chapter 3

- a. $19.992 \times 0.9048 + 20.993 \times 0.0027 + 21.991 \times 0.0925 = 20.18$ amu
- b. There are three different forms of Br_2 with masses of 157.84, 159.84, and 161.84 amu. The middle form is the most abundant because it has the largest peak, and the other two occur in smaller but similar amounts.
- c. An empirical formula shows the simplified ratio of atoms in a compound (for example, CH_2O) while a molecular formula shows the exact number of atoms in a molecule ($\text{C}_6\text{H}_{12}\text{O}_6$).
- d. Equations need to be balanced to perform calculations and to make sure that the law of conservation of mass is followed.
- e. aqueous (aq), solid (s), liquid (l), gas (g)
- f. The ratio of moles in one substance to moles of another in a balanced equation
- g. The reactant that limits the amount of product that can be made (and is used up completely)
- h. A mixture of ideal proportion where reactants are used completely
- i. The theoretical amount of reactant that is expected to be produced
- j. The percent yield is the percent of the theoretical yield that was actually produced in the experiment. The formula is

$$\text{percent yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

k.

$$\begin{aligned}\frac{0.2 \text{ g}}{1 \text{ carat}} \times 2.5 \text{ carats} &= 0.5 \text{ g} \\ 0.5 \text{ g} \times \frac{1 \text{ mol}}{12.011 \text{ g}} &= 0.0416 \text{ mol} \\ 0.0416 \text{ mol} \times 6.022 \times 10^{23} \text{ atoms/mol} &\approx \boxed{2.507 \times 10^{22} \text{ atoms}}\end{aligned}$$

l. i.

Molar mass of $\text{Fe}_2\text{O}_3 = 159.687 \text{ g/mol}$

$$150 \text{ g} = \frac{1 \text{ mol}}{159.687 \text{ g}} \approx \boxed{0.94 \text{ mol}}$$

ii.

Molar mass of $\text{NO}_2 = 46.01 \text{ g/mol}$

$$10 \text{ mg} \times \frac{1 \text{ g}}{1000 \text{ mg}} \times \frac{1 \text{ mol}}{46.01 \text{ g}} \approx \boxed{0.000217 \text{ mol}}$$

iii.

$$1.5 \times 10^{16} \text{ molecules} \times \frac{1 \text{ mol}}{6.022 \times 10^{23} \text{ molecules}} \approx \boxed{2.5 \times 10^{-1} \text{ mol}}$$

- m. i. Molar masses: $\text{CH}_2\text{O} = 30.026 \text{ g/mol}$; $\text{C} = 12.011 \text{ g/mol}$; $2\text{H} = 2.016 \text{ g/mol}$;
 $\text{O} = 16 \text{ g/mol}$

$$\frac{12.011}{30.026} = 40.00\% \text{ C}$$

$$\frac{2.016}{30.026} = 6.71\% \text{ H}$$

$$\frac{16}{30.026} = 53.29\% \text{ O}$$

ii. same as i. same empirical formula

iii. same as i. and ii. also same empirical formula

- n. Compound is 48.64% C (12.011 g/mol), 8.16% H (1.008 g/mol), 43.2% O (16 g/mol).
 Assuming a mass of 100g—

$$48.64 \text{ g C} \times \frac{1 \text{ mol}}{12.011 \text{ g C}} = 4.05 \text{ mol C}$$

$$8.16 \text{ g H} \times \frac{1 \text{ mol}}{1.008 \text{ g H}} = 8.10 \text{ mol H}$$

$$43.2 \text{ g O} \times \frac{1 \text{ mol}}{16 \text{ g O}} = 2.70 \text{ mol O}$$

Mole ratio:

$$4.05/2.70 = 1.5$$

$$8.10/2.70 = 3.0$$

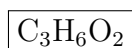
$$2.70/2.70 = 1.0$$

As whole numbers:

$$1.5 \times 2 = 3$$

$$3.0 \times 2 = 6$$

$$1.0 \times 2 = 2$$



- o. 69.6% sulfur (32.07 g/mol); 30.4% nitrogen (14.01 g/mol); molar mass 184 g/mol
 Empirical:

$$69.6 \text{ g S} \times \frac{1 \text{ mol}}{32.07 \text{ g}} = 2.17 \text{ mol S}$$

$$30.4 \text{ g N} \times \frac{1 \text{ mol}}{14.01 \text{ g}} = 2.17 \text{ mol N}$$

$$\boxed{\text{SN}}$$

Molecular:

$$0.696 \times \frac{184 \text{ g}}{1 \text{ mol}} \times \frac{1 \text{ mol}}{32.07 \text{ g}} = 4$$

$$0.304 \times \frac{184 \text{ g}}{1 \text{ mol}} \times \frac{1 \text{ mol}}{14.01 \text{ g}} = 4$$

$$\boxed{\text{S}_4\text{N}_4}$$

- p.
- $6 \text{ KO}_{2(\text{s})} + 4 \text{ H}_2\text{O}_{(\text{l})} \rightarrow 6 \text{ KOH}_{(\text{aq})} + 4 \text{ O}_{2(\text{g})} + \text{H}_2\text{O}_{2(\text{aq})}$
 - $\text{Fe}_2\text{O}_{3(\text{s})} + 6 \text{ HNO}_{3(\text{aq})} \rightarrow 2 \text{ Fe}(\text{NO}_3)_{3(\text{aq})} + 3 \text{ H}_2\text{O}_{(\text{l})}$
 - $4 \text{ NH}_{3(\text{g})} + 5 \text{ O}_{2(\text{g})} \rightarrow 4 \text{ NO}_{(\text{g})} + 6 \text{ H}_2\text{O}_{(\text{g})}$
 - $\text{PCl}_{5(\text{l})} + 4 \text{ H}_2\text{O}_{(\text{l})} \rightarrow \text{H}_3\text{PO}_{4(\text{aq})} + 5 \text{ HCl}_{(\text{g})}$
 - $2 \text{ CaO}_{(\text{s})} + 5 \text{ C}_{(\text{s})} \rightarrow 2 \text{ CaC}_{2(\text{s})} + \text{CO}_{2(\text{g})}$
 - $6 \text{ MoS}_{2(\text{s})} + 21 \text{ O}_{2(\text{g})} \rightarrow 6 \text{ MoO}_{3(\text{s})} + 12 \text{ SO}_{2(\text{g})}$
 - $\text{FeCO}_{3(\text{s})} + \text{H}_2\text{CO}_{3(\text{aq})} \rightarrow \text{Fe}(\text{HCO}_3)_{2(\text{aq})}$

q. Equation: $\text{N}_{2(\text{g})} + 3 \text{ H}_{2(\text{g})} \rightarrow 2 \text{ NH}_{3(\text{g})}$

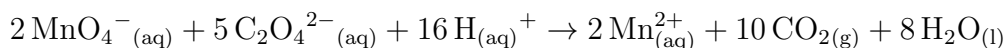
Molar mass of $\text{N}_2 = 28.014 \text{ g/mol}$; molar mass of $\text{H}_2 = 2.016 \text{ g/mol}$. $1000 \text{ g}/28.014 \text{ g/mol} = 35.70 \text{ mol N}_2$ and $500 \text{ g}/2.016 \text{ g/mol} = 248.02 \text{ mol H}_2$ are available. $248.02/3 = 82.67$; N_2 is the limiting reagent

$$35.70 \text{ mol N}_2 \times \frac{2 \text{ mol NH}_3}{1 \text{ mol N}_2} \times \frac{17.034 \text{ g NH}_3}{1 \text{ mol NH}_3} \approx \boxed{1216.5 \text{ g NH}_3}$$

$$35.70 \text{ mol N}_2 \times \frac{3 \text{ mol H}_2}{1 \text{ mol N}_2} = 107.1 \text{ mol H}_2 \text{ needed for reaction}$$

$$248.02 - 107.1 = 140.92 \text{ mol H}_2 \times \frac{2.016 \text{ g}}{1 \text{ mol H}_2} = 284.09 \text{ g H}_2 \text{ left over}$$

r.



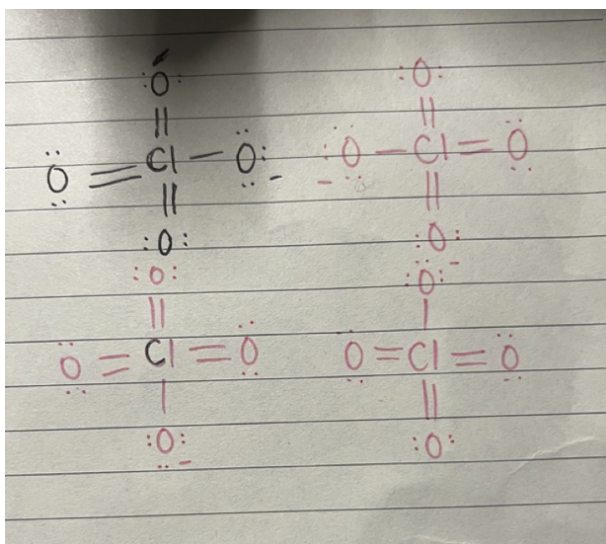
$(0.20 \text{ mol/L})(0.020 \text{ L}) = 0.0040 \text{ mol MnO}_4^-$ and $(0.10 \text{ mol/L})(0.050 \text{ L}) = 0.0050 \text{ mol C}_2\text{O}_4^{2-}$ available. $0.0040 \times \frac{5}{2} = 0.010 \text{ mol C}_2\text{O}_4^{2-}$ needed to react with all of the MnO_4^- ; lim reagent is $\text{C}_2\text{O}_4^{2-}$

$$0.0050 \text{ mol C}_2\text{O}_4^{2-} \times \frac{10 \text{ mol CO}_2}{5 \text{ mol C}_2\text{O}_4^{2-}} = \boxed{0.010 \text{ mol CO}_2}$$

7 Additional knowledge

a. $1s^2 2s^2 2p^6$

Kr $5s^2 4d^{10} 5p^2$



- b.
- c. sp^2
- d. Trigonal planar; 120 degrees

8 Questions for review

Question 6a. had some trouble understanding the table; 6b I don't remember doing in honors chem so I had help from previous AP Chem students on those questions. 7e. I remembered but would like to review VSEPR in more detail; will do before school starts